

UniCore Prototype Specification

Architecture & PL/SQL Implementation

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Abstract

The UniCore Prototype is a 3-tier software platform engineered to demonstrate real-time transaction safeguards, PL/SQL trigger automation, role-based authentication, and immutable audit logging.

1 Implemented PL/SQL Stored Procedures

Listing 1: Atomic Hostel Allotment Stored Procedure

```
CREATE OR REPLACE PROCEDURE hostel.hostel_allot(  
    p_student_id UUID,  
    p_room_id UUID,  
    p_academic_year TEXT  
) AS $$  
DECLARE  
    v_bed_count INT;  
    v_current_alloc INT;  
BEGIN  
    SELECT bed_count INTO v_bed_count  
    FROM hostel.hostel_rooms  
    WHERE id = p_room_id FOR UPDATE;  
  
    SELECT COUNT(*) INTO v_current_alloc  
    FROM hostel.allocations  
    WHERE room_id = p_room_id AND status = 'active';  
  
    IF v_current_alloc >= v_bed_count THEN  
        RAISE EXCEPTION 'Room capacity reached.';  
    END IF;  
  
    INSERT INTO hostel.allocations (student_id, room_id, academic_year,  
        status)  
    VALUES (p_student_id, p_room_id, p_academic_year, 'active');  
END;  
$$ LANGUAGE plpgsql;
```